The Collected Mathematical Papers of Arthur Cayley
Vol. XII

Book Pages 101–125 (Papers 819 concl.–825 Chs. I–II)

Arthur Cayley (1821–1895)

Published 1889

Transcribed from the original Cambridge University Press edition, 1889.
This file covers book pages 101–125; restored from scan to correct prior errors.

Contents

819. On Two Cases of the Quadric Transformation between Two Planes (concluded)

[p. 101] Writing for convenience (a, b, c) for the coordinates of the fixed point, and (x_1, y_1, z_1), (x_2, y_2, z_2) for those of the other two points, the formulæ with A, B, C give thus the correspondence

$$x_2 : y_2 : z_2 \;=\; bcx_1^2 - a^2y_1z_1 \;:\; ca\,y_1^2 - b^2z_1x_1 \;:\; ab\,z_1^2 - c^2x_1y_1,$$

which is the first of the two cases in question. These equations give reciprocally

$$x_1 : y_1 : z_1 \;=\; bc\,x_2^2 - a^2y_2z_2 \;:\; ca\,y_2^2 - b^2z_2x_2 \;:\; ab\,z_2^2 - c^2x_2y_2,$$

or the correspondence is a (1, 1) quadric correspondence.

The formulæ with P, Q, R give in like manner

$$x_2 : y_2 : z_2 = a(ax_1^2 + by_1^2 + cz_1^2) - a_1(a''\,y_1z_1 + b'\,z_1x_1 + c'\,x_1y_1),\ \text{\&c.},$$

or if for shortness

$$\Omega_1 = ax_1^2 + by_1^2 + cz_1^2, \qquad \Theta_1 = a''\,y_1z_1 + b'\,z_1x_1 + c'\,x_1y_1,$$

then

$$x_2 : y_2 : z_2 \;=\; a\Omega_1 + a_1\Theta_1 \;:\; b\Omega_1 + b_1\Theta_1 \;:\; c\Omega_1 + c_1\Theta_1,$$

which is the second of the two cases. We have reciprocally

$$x_1 : y_1 : z_1 \;=\; a\Omega_2 + a_1\Theta_2 \;:\; b\Omega_2 + b_1\Theta_2 \;:\; c\Omega_2 + c_1\Theta_2,$$

where

$$\Omega_2 = ax_2^2 + by_2^2 + cz_2^2, \qquad \Theta_2 = a''\,y_2z_2 + b'\,z_2x_2 + c'\,x_2y_2,$$

and the correspondence is thus in this case also a (1, 1) quadric correspondence.

820. On a Problem of Analytical Geometry

[From the *Johns Hopkins University Circulars*, No. 15 (1882), p. 209.]

[p. 102] THE object of the present note is only to call attention to a problem of Analytical Geometry which presents itself in connexion with the reduction of an algebraical integral, and which is solved, pp. 21, 22 of Clebsch and Gordan's *Theorie der Abel'schen Functionen* (Leipzig, 1866); viz. the problem is, considering a curve $f = 0$ of the order n, to find the conic $\Omega = 0$ of the order $n-2$ passing through the remaining $n-2$ points of intersection of the line with the curve f, through the double points of f, and through as many other given points as are required for the determination of the curve. If, for instance, f is a quartic curve without double points, then Ω is the quadric curve which passes through the remaining two intersections of the line with Ω, and through three given points. Take $(\xi,\,\eta,\,\zeta)$, $(\xi_1,\,\eta_1,\,\zeta_1)$ the coordinates of the two given points on the curve f (f is, say of the order $n = 4$); $\theta(x,\,y,\,z) = 0$ the equation of the required curve. By the equation of suppositions originally of the form, that the line $x = x' + \rho x''$, &c. joining $(x',\,y',\,z')$, $(x'',\,y'',\,z'')$ meets the curve θ where $\theta = 0$ in two points, the equation in ρ originally of the fourth order in $(\lambda,\,\mu)$ but which divides by $\lambda\mu$, and which when this factor is thrown out becomes

$$\alpha\lambda^2 + 2\beta\lambda\mu + \gamma\mu^2 = 0;$$

where

$$\alpha = (\xi_1,\,\eta_1,\,\zeta_1)^2\,\theta, \quad \beta = (\xi_1,\,\eta_1,\,\zeta_1)(\xi,\,\eta,\,\zeta)\,\theta, \quad \gamma = (\xi,\,\eta,\,\zeta)^2\,\theta,$$

where for shortness f_1, f are written to denote $f(\xi_1,\,\eta_1,\,\zeta_1)$, $f(\xi,\,\eta,\,\zeta)$ respectively.

[p. 103] The condition as to the two points obviously is that, making the same substitution $x,\,y,\,z = \lambda\xi_1 + \mu\xi,\ \lambda\eta_1 + \mu\eta,\ \lambda\zeta_1 + \mu\zeta$ in the equation $\Omega = (a,\ldots\,)(x,\,y,\,z)^2 = 0$, we must obtain the same quadric equation in $\lambda : \mu$. We have thus two conditions, which, introducing an indeterminate multiplier θ, are expressed by the three equations

$$\begin{aligned}(a,\ldots\,)(\xi_1,\,\eta_1,\,\zeta_1)^2 &= \theta\alpha,\\ (a,\ldots\,)(\xi_1,\,\eta_1,\,\zeta_1)(\xi,\,\eta,\,\zeta) &= \theta\beta,\\ (a,\ldots\,)(\xi,\,\eta,\,\zeta)^2 &= \theta\gamma.\end{aligned}$$

The conditions as to the three points are obviously

$$\begin{aligned}(a,\ldots\,)(x_1,\,y_1,\,z_1)^2 &= 0,\\ (a,\ldots\,)(x_2,\,y_2,\,z_2)^2 &= 0,\\ (a,\ldots\,)(x_3,\,y_3,\,z_3)^2 &= 0,\end{aligned}$$

and these equations determine the ratios of $a,\,b,\,c,\,f,\,g,\,h$. But to complete the solution the convenient course is to regard the function $\Omega, = (a,\ldots\,)(x,\,y,\,z)^2$ as a quantity to be determined, and consequently to join to the foregoing the equation

$$(a,\ldots\,)(x,\,y,\,z)^2 = \Omega;$$

we have thus seven equations from which $(a,\,b,\,c,\,f,\,g,\,h)$ may be eliminated, the result being expressed by means of a determinant of the seventh order

$$\begin{vmatrix} (x,\,y,\,z)^2 & \Omega \\ (\xi_1,\,\eta_1,\,\zeta_1)^2 & \theta\alpha \\ (\xi_1,\,\eta_1,\,\zeta_1)(\xi,\,\eta,\,\zeta) & \theta\beta \\ (\xi,\,\eta,\,\zeta)^2 & \theta\gamma \\ (x_1,\,y_1,\,z_1)^2 & 0 \\ (x_2,\,y_2,\,z_2)^2 & 0 \\ (x_3,\,y_3,\,z_3)^2 & 0 \end{vmatrix} = 0,$$

viz. this is an equation of the form $A\Omega = \theta V$, where A is a constant determinant of the sixth order (i.e. a determinant not involving x, y, z), V a determinant of the seventh order, a quadric function of (x, y, z), obtained from the foregoing determinant by writing therein $\Omega = 0$ and $\theta = 1$; the multiplier θ is and remains arbitrary; but it is convenient to take it to be $= 1$, viz. we have then not only find the equation $\Omega = 0$ of the required conic, but we put a determinate value for Ω, say $\Omega = A^{-1}V$, which is, in fact $(a, \dots\ )(x, y, z)^2$, viz. this gives $a = (\xi_1\beta_1+\eta_1\delta_1+\zeta_1\theta_1)f_1$, and so on for $(x, y, z) = (\xi, \eta, \zeta)$, we have

$$\Omega_1 = \gamma_1 = (\xi_1\beta_1 + \eta_1\delta_1 + \zeta_1\theta_1)f_1.$$

821. On the Geometrical Representation of an Equation between Two Variables

[From the *Johns Hopkins University Circulars*, No. 15 (1882), p. 210.]

[p. 104] AN equation between two variables cannot be represented in a satisfactory manner by a curve, for this serves only to represent the corresponding real values of the variables; to represent the imaginary values the natural course is to represent each variable by a point in a plane, viz. the variable $z, = x + iy$, will be represented by a point the coordinates of which are the components x and y of the variable, and similarly the variable $z', = x' + iy'$, by a point the coordinates of which are the components x' and y' of the variable; the equation between the two variables will then establish a correspondence between the two variables points respectively, or say a correspondence between the planes which contain the two points respectively: and it is this correspondence of two planes which is the proper geometrical representation of the equation between the two variables; to exhibit the correspondence we may in either of the planes draw a network of curves at pleasure, and then draw in the other plane the network of corresponding curves. This well-known theory [can be] illustrated by the case $z' = \sqrt{z^2 - 1}$: taking in the infinite half-plane y positive about the origin as centre a system of semi-circles, to these correspond in the infinite plane of $x'y'$ a series of lemniscate-shaped curves: and by means of these it is easy to show in the second plane the path corresponding to a given path of the point $z, = x + iy$, in the first half-plane.

822. On Associative Imaginaries

[From the *Johns Hopkins University Circulars*, No. 15 (1882), pp. 211, 212.]

[p. 105] THE imaginaries (x, y) defined by the equations

$$\begin{aligned} x^2 &= ax + by, \\ xy &= cx + dy, \\ yx &= ex + fy, \\ y^2 &= gx + hy, \end{aligned}$$

will not be in general associative; to make them so, we must have 8 double relations corresponding to the combinations $x^2.x$, $xy.x$, $yx.x$, $x^2.y$, $xy.y$, $yx.y$, $y^2.x$, $y^2.y$ respectively, viz. the first of these gives $x.x.x = x.x.x$, that is, $0 = x(ax+by)-(ax+by)x$, $= b(xy-yx)$, $= b\{(c-e)x+(d-f)y\}$; and similarly for each of the other terms. We thus obtain apparently 16, but really only 12, relations between the 8 coefficients a, b, c, d, e, f, g, h, viz. the relations so obtained are

$$\begin{array}{lll} b(c-e)=0 \text{ (twice)}, & b(d-f)=0, & g(c-e)=0, \quad g(d-f)=0 \text{ (twice)} \\ bg-cd=0 \text{ (twice)}, & bg-ef=0 \text{ (twice)}, & \\ c^2+dg-cg-eh=0, & d^2+bc-cd-af+bg-ah=0, & \\ a(c-e)+de-cf=0, & h(f-d)-cf+de=0. & \end{array}$$

From the first four equations it appears that either $b=0$ and $g=0$, or else $c=e$ and $d=f$: the latter solution may be set aside for the moment, giving the commutative system

$$\begin{aligned} x^2 &= ax + by, \\ xy = yx &= cx + dy, \\ y^2 &= gx + hy. \end{aligned}$$

[p. 106] In order that this may be associative, we must still have the relations

$$\begin{aligned} bg - cd &= 0, \\ c^2 + dg - cg - ch &= 0, \\ d^2 + bc - ad - bh &= 0, \end{aligned}$$

which are all three of them satisfied by $g = \dfrac{cd}{b}$, $h = \dfrac{d^2+bc-ad}{b}$, viz. we thus have the associative and commutative system

$$\begin{aligned} x^2 &= ax + by, \\ xy = yx &= cx + dy, \\ y^2 &= \frac{cd}{b}x + \frac{d^2+bc-ad}{b}y. \end{aligned}$$

I did not perceive how to identify this system with any of the double algebras of B. Peirce's *Linear Associative Algebra*, see pp. 120–122 of the Reprint, *American Journal of Mathematics*, t. iv. (1881); but it has been pointed out to me by Mr C. S. Peirce that my system, in the general case $ad-bc \neq 0$, is reducible to the form (a_1), see p. 122 (l.c.). The object of the present Note is to exhibit in the simple case of two imaginaries the whole system of relations which must subsist between the coefficients in order that the imaginaries may be associative; that is, the system of equations which are solved implicitly by the establishment of the several multiplication tables of the memoir just referred to.

823. On the Geometrical Interpretation of Certain Formulæ in Elliptic Functions

[From the *Johns Hopkins University Circulars*, No. 17 (1882), p. 238.]

[p. 107] I HAVE given in my *Elliptic Functions* expressions for the sn of $u+\tfrac{1}{2}K$, $u+\tfrac{1}{2}iK'$, $u+\tfrac{1}{2}K+\tfrac{1}{2}iK'$; but it is better to consider the dn, sn, cn of these combinations respectively, and to write the formulæ thus

$$\mathrm{dn}^2(u+\tfrac{1}{2}K)=k'\,\frac{\mathrm{dn}\,u-(1-k')\,\mathrm{sn}\,u\,\mathrm{cn}\,u}{\mathrm{dn}\,u+(1-k')\,\mathrm{sn}\,u\,\mathrm{cn}\,u}\,,\quad =k'\,\frac{1-k'x-(1-k')y}{1-k'x+(1-k')y}\,;$$

$$\mathrm{sn}^2(u+\tfrac{1}{2}iK')=\frac{1}{k}\,\frac{(1+k)\,\mathrm{sn}\,u+i\,\mathrm{cn}\,u\,\mathrm{dn}\,u}{(1+k)\,\mathrm{sn}\,u-i\,\mathrm{cn}\,u\,\mathrm{dn}\,u}\,,\quad =\frac{1}{k}\,\frac{(1+k)x+iy}{(1+k)x-iy}\,;$$

$$\mathrm{cn}^2(u+\tfrac{1}{2}K+\tfrac{1}{2}iK')=-\,\frac{ik'}{k}\,\frac{\mathrm{cn}\,u-(k+ik')\,\mathrm{sn}\,u\,\mathrm{dn}\,u}{\mathrm{cn}\,u+(k+ik')\,\mathrm{sn}\,u\,\mathrm{dn}\,u}\,,\quad =-\,\frac{ik'}{k}\,\frac{1-x-(k+ik')y}{1-x+(k+ik')y}\,,$$

where in the last set of ratios x, y are used to denote $\mathrm{sn}\,u\,\mathrm{cn}\,u$ and $\mathrm{sn}\,u\,\mathrm{dn}\,u$ respectively; and the three combinations above show that for u are written respectively the values just mentioned. The curve $y^2=(1-x^2)(1-k^2x^2)$ has an inflexion at infinity on the line $x=0$; and the three tangents from the inflexion are at the points $x,\ y=0,\ 1,\ 0$ and $\left(\frac{1}{k},\ 0\right)$ respectively; touching the curve at the points $x,\ y=(0,\ 0),\ (1,\ 0),\ \left(\frac{1}{k},\ 0\right)$ respectively; hence these points are sextactic points.

[p. 108] We may from any one of these, for instance the point $(0,\ 0)$, draw four tangents to the curve, $(1+k)x+iy=0$, $(1+k)x-iy=0$; $(1-k)x+iy=0$, $(1-k)x-iy=0$; where the first and second of these lines form a pair, and the third and fourth of them form a pair, viz. the two tangents of a pair touch in points such that the line joining them passes through the point of inflexion; in particular, for the first-mentioned pair, the equation of the line joining the points of contact is $1+kx=0$. The tangents from belonging to a pair of tangents are precisely those which present themselves in the formulæ; thus if $T_1=(1+k)x+iy$, $T_2=(1+k)x-iy$, the second of the three formulæ is $\mathrm{sn}^2(u+\tfrac{1}{2}iK')=\frac{1}{k}\,\frac{T_1}{T_2}$; and the other two formulæ correspond in like manner to pairs of tangents from the sextactic points $\left(\frac{1}{k},\ 0\right)$, and $(1,\ 0)$ respectively. The formulæ are connected with the fundamental equations expressing the functions sn, cn, dn as quotients of theta-functions.

824. Note on the Formulæ of Trigonometry

[From the *Johns Hopkins University Circulars*, No. 17 (1882), p. 241.]

[p. 108] THE equations $a = c\cos B + b\cos C$, $b = c\cos A + a\cos C$, $c = b\cos A + a\cos B$, which connect together the sides a, b, c and the angles A, B, C of a plane triangle, may be presented in an algebraical rational form, by introducing in place of the angles A, B, C the functions $\cos A + i\sin A$, $\cos B + i\sin B$, $\cos C + i\sin C$, viz. calling these $\frac{x}{x'}$, $\frac{y}{y'}$, $\frac{z}{z'}$ respectively, or, what is the same thing, writing $2\cos A = \frac{x}{x'} + \frac{x'}{x}$, $2\cos B = \frac{y}{y'} + \frac{y'}{y}$, $2\cos C = \frac{z}{z'} + \frac{z'}{z}$, then the foregoing equations may be written

$$\begin{aligned}
(-2a)xx' \;+\; y(zy' + z'y)b \;+\; z(xy' + x'y)c &= 0,\\
x(yz' + y'z)a + (-2b)yy' + z(xy' + x'y)c &= 0,\\
x(yz' + y'z)a + y(zx' + z'x)b + (-2c)zz' &= 0,
\end{aligned}$$

that is, as a system of bipartite equations linear in (a, b, c) and cubic in (x, y, z, x', y', z') respectively.

Similarly in Spherical Trigonometry, writing as above for the angles, and for the sides writing in like manner $2\cos a = \frac{a}{\delta} + \frac{\delta}{a}$, $2\cos b = \frac{\beta}{\delta} + \frac{\delta}{\beta}$, $2\cos c = \frac{\gamma}{\delta} + \frac{\delta}{\gamma}$, we have a system of bipartite equations separately homogeneous in regard to (x, y, z, x', y', z') and $(\alpha, \beta, \gamma, \delta)$ respectively.

825. A Memoir on the Abelian and Theta Functions

[Chapters I to III, *American Journal of Mathematics*, t. v. (1882), pp. 137–179;
Chapters IV to VII, ib., t. vii. (1885), pp. 101–167.]

[p. 109] THE present memoir is based upon Clebsch and Gordan's *Theorie der Abel'schen Functionen*, Leipzig, 1866 (here cited as C. and G.); the employment of differential rather than of integral equations is a novelty; but the chief addition to the theory consists in the determination which I have made for the cubic curve, and also (but not as yet in a perfect form) for the quartic curve, of the differential expression $d\Pi_{12}$ (or as I have it $d\Pi$) in the integral of the third kind $\int_a^b d\Pi_{12}$ in the final normal form (endliche Normalform) for which we have (p. 117) $\int_a^b d\Pi_{12} = \int_1^2 d\Pi_{ab}$, the limits and parametric points interchangeable. The want of this determination presented itself to me as a locus in the theory during the course of lectures on the subject which I had the pleasure of giving in the Johns Hopkins University, Baltimore, U.S.A., in the months January to June, 1882, and I succeeded in effecting it for the cubic curve; but it was not until shortly after my return to England that I was able partially to effect the like determination in the far more difficult case of the quartic curve. The memoir contains, with additional developments, a reproduction of the course of lectures just referred to. I have endeavoured to simplify as much as possible the notations and demonstrations of Clebsch and Gordan's admirable treatise; to bring some of the geometrical results into greater prominence; and to illustrate the theory by examples in regard to the cubic, the nodal quartic, and the general quartic curves respectively. The various chapters are: I, Abel's Theorem; II, Proof of Abel's Theorem; III, The Major Function (continued); IV, The Major Function (continued); V, Miscellaneous Investigations; VI, The Nodal Quartic; VII, The Functions T, U, V, Θ. The paragraphs of the whole memoir will be numbered continuously.

Chapter I. Abel's Theorem.

The Differential Pure and Affected Theorems. Art. Nos. 1 to 5.
[p. 110] 1. We have a fixed curve and a variable curve, and the differential pure theorem consists in a set of linear relations between the displacements of the intersections of the two curves; in the affected theorem, a linear function of the displacements is equated to another differential expression. I state the two theorems, giving afterwards the necessary explanations.

The pure theorem is

$$\Sigma\,(x,\,y,\,z)^{n-3}\,du = 0.$$

The affected theorem is

$$\Sigma\,\frac{(x,\,y,\,z)^{n-3}\,du}{012} \;=\; -\,\frac{\delta\phi_1}{\phi_1}\,\frac{\delta\phi_2}{\phi_2}\;*\;.$$

2. We have a fixed curve $f=0$, or say the curve f, or simply the fixed curve, of the order n, with δ dps, and therefore of the deficiency $\frac{1}{2}(n-1)(n-2)-\delta, = p$. The expression "the dps" means always the δ dps of f.

And we have a variable curve $\phi=0$, or say the curve ϕ, or simply the variable curve, of the order m, passing through the dps and besides meeting the fixed curve in $mn-2\delta$ variable points.

Moreover, du is the displacement of the current point 0, coordinates $(x,\,y,\,z)$, on the fixed curve, viz. the equation $f=0$ being satisfied, we have

$$\frac{df}{dx}\,dx+\frac{df}{dy}\,dy+\frac{df}{dz}\,dz=0,$$

and we thence have

$$\frac{df}{dx} : \frac{df}{dy} : \frac{df}{dz} \;=\; y\,dz - z\,dy : z\,dx - x\,dz : x\,dy - y\,dx,$$

so that we have three equal values each of which is put $= du$, viz. we write

$$\frac{y\,dz - z\,dy}{\dfrac{df}{dx}} \;=\; \frac{z\,dx - x\,dz}{\dfrac{df}{dy}} \;=\; \frac{x\,dy - y\,dx}{\dfrac{df}{dz}} \;=\; du,$$

and du as thus defined is the displacement.

[p. 111] 3. $(x,\,y,\,z)^{n-3} = 0$ is the minor curve, viz. the general curve of the order $n-3$ which passes through the dps; and the function $(x,\,y,\,z)^{n-3}$ in the proper major function, viz. the implicit factor of the function is so determined that, taking $0 = 1$, the current point at 1, that is, writing $(x_1,\,y_1,\,z_1)$ for $(x,\,y,\,z)$, the function $(x,\,y,\,z)^{n-3}$ reduces itself to the polar function $\left(x_1\dfrac{d}{dx_2} + y_1\dfrac{d}{dy_2} + z_1\dfrac{d}{dz_2}\right) f_2$, afterwards written $n.\,1^{n-1}\,2$, of f: this implies that taking $0 = 2$, the current point at 2, the function reduces itself to the polar function $n.\,1\,2^{n-1}$.

ϕ_1 is what ϕ becomes on writing therein $(x_1,\,y_1,\,z_1)$ for $(x,\,y,\,z)$; and similarly ϕ_2 is what ϕ becomes on writing therein $(x_2,\,y_2,\,z_2)$ for $(x,\,y,\,z)$.

δ denotes differentiation in regard only to the coefficients of ϕ; viz. writing $\phi = (a,\dots\,)(x,\,y,\,z)^m$ we have $\delta\phi = (da,\dots\,)(x,\,y,\,z)^m$, and similarly $\delta\phi_1$ and $\delta\phi_2 = (da,\dots\,)(x_1,\,y_1,\,z_1)^m$ and $(da,\dots\,)(x_2,\,y_2,\,z_2)^m$ respectively.

The sum Σ extends to all the variable intersections of the two curves.

3. As to the meaning of the theorems, consider first the pure theorem. The variable intersections are not all of them arbitrary points on the fixed curve; a certain number of them taken at pleasure on the fixed curve will determine the remaining variable intersections; and there are thus a certain number of integral relations between the coordinates of the variable intersections; to each such integral relation there corresponds a linear relation between the displacements du of these points, or say a displacement-relation; and it is precisely these displacement-relations which are given by the theorem, viz. the equation

$$\Sigma\,(x,\,y,\,z)^{n-3}\,du = 0$$

breaks up into as many linear relations as there are arbitrary constants in the function $(x,\,y,\,z)^{n-3}$, which equated to zero gives a curve of the order $n-3$ passing through the dps; for instance, if there be no singularities $\delta = 0$, then the relation gives the single relation $\Sigma\,du = 0$; but $n = 4$, $\delta = 0$, the equation gives the three relations $\Sigma x\,du = 0$, $\Sigma y\,du = 0$, $\Sigma z\,du = 0$.

[p. 112] 4. It is of course important to show, and it will be shown, that the number of independent displacement-relations given by the theorem is equal to the number of independent integral relations between the variable intersections.

5. Observe that the pure theorem gives all the displacement-relations between the variable intersections; the coefficients of ϕ are hereby led to see the nature of the affected theorem. The variable intersections, the coefficients of ϕ are thereby determined in terms of the coordinates of the assumed variable intersections*, and hence the value of $-\dfrac{\delta\phi_1}{\phi_1}\dfrac{\delta\phi_2}{\phi_2}$ is given as a linear function of the corresponding displacements du; and, substituting this value, the affected theorem gives a linear relation between the displacements du of the several variable intersections. But any such linear relation must clearly be a mere linear combination of the displacement-relations $\Sigma\,(x,\,y,\,z)^{n-3}\,du = 0$ given by the pure theorem.

Examples of the Pure Theorem—The Fixed Curve a Cubic. Art. Nos. 6 to 12.

6. The pure theorem is not applicable to the case $n = 2$, the fixed curve a conic: it in fact gives no displacement-relation; and this is as it should be, for the variable intersections are all of them arbitrary.

The next case is $n = 3$, $\delta = 0$, the fixed curve a cubic. For greater simplicity the equation is taken in Cartesian coordinates. In general for such an equation, writing in the homogeneous formula $z = 1$, we have

$$du = \frac{dx}{-\dfrac{df}{dy}} = \frac{dy}{\dfrac{df}{dx}},$$

the two values being of course equal in virtue of $\dfrac{df}{dx}\,dx + \dfrac{df}{dy}\,dy = 0$; taking the former value and considering $\dfrac{df}{dy}$ as expressed in terms of x, let this be called P (of course, P is an irrational function of x); then we have $du = \dfrac{dx}{P}$; and similarly $du_1 = \dfrac{dx_1}{P_1}$, &c.

The fixed curve being then a cubic, let the variable curve be a line; this meets the cubic in three points, say 1, 2, 3; and any two of these determine the line, and therefore the third point; there should therefore be one integral relation, and consequently one displacement-relation; and this is in fact the case by the theorem, viz. we have $\Sigma du = 0$, or, $du_1 + du_2 + du_3 = 0$, that is in the same thing,

$$\frac{dx_1}{P_1} + \frac{dx_2}{P_2} + \frac{dx_3}{P_3} = 0.$$

[p. 113] The corresponding integral equation is the equation which expresses that the points 1, 2, 3 are in a line, viz. considering y_1, y_2, y_3 as given functions of x_1, x_2, x_3 respectively, this is

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0,$$

or, in the notation already made use of for such a determinant, $123 = 0$.

7. This equation $du_1 + du_2 + du_3 = 0$, here $du = \dfrac{dx}{P}$, has a peculiar interpretation when we consider the coefficients of the cubic as arbitrary constants, and therefore the cubic curve, we in effect determine y_1 as a function of x_1 and the constants; the points 1 and 2 determine the line and the third point; we have thus x_3, y_3 determined as a function of x_1, x_2 and the constants.

Considering x_3 as thus expressed, we have $dx_3 = \dfrac{dx_3}{dx_1}\,dx_1 + \dfrac{dx_3}{dx_2}\,dx_2$, an equation which must agree with $du_1 + du_2 + du_3 = 0$, that is, with $dx_3 = -\dfrac{P_3}{P_1}\,dx_1 - \dfrac{P_3}{P_2}\,dx_2$. It follows that we have $\dfrac{dx_3}{dx_1} : \dfrac{dx_3}{dx_2} = P_1 : P_2$, and taking the logarithms and differentiating with regard to x_1 and x_2, we find $\dfrac{d}{dx_1}\dfrac{d}{dx_2}\log\left(\dfrac{dx_3}{dx_1} : \dfrac{dx_3}{dx_2}\right) = 0$, a partial differential equation of the third order, independent of any particular cubic curve, and satisfied by x_3 considered as a function of x_1, x_2 and the nine constants. Observe that starting from the expression for x_3, and proceeding to the differential coefficients of the third order, we have ten equations from which the nine constants can be eliminated, that is, we ought to have a partial differential equation of the third order; and conversely that the equation for x_3, as containing nine arbitrary constants, is a complete solution of the partial differential equation: the complete solution of the partial differential equation in question is thus the equation which expresses that 3 is the remaining intersection of the line through 1 and 2 with a cubic.

8. As a particular illustration, we have $\dfrac{dx}{P} = du$, $\Sigma du = 0$, and the integral relation $123 = 0$. I give two new ways of obtaining this—first by differentiation.

We have the equation $123 = 0$. I give a direct verification.

To find $\dfrac{dx_3}{dx_1}$, $\dfrac{dx_3}{dx_2}$, where 1, 2, 3 are on the cubic, $f = 0$. Let ϕ be any conic, $\phi = 0$, taken so that $\dfrac{d\phi}{dx} \div f = du$, and as the equation of ϕ at 0 of the curve, and it may be regarded as the determination of ϕ; with the dps and the ϕ other points $a, \dots, h$ on the curve.

This being so, considering as variable the original coordinates f, we have by differentiation

$$\Sigma\, \frac{1}{P}\, \delta \log\, \frac{f}{(ax^3 + by^3 + cz^3 + 6lxyz)} = 0,$$

or what is the same, taking the logarithms and differentiating with regard to one of the constants a, the equation $\Sigma du = 0$ is satisfied.

We have shown also that the cubic curve $f = 0$, and the cube derived line $\phi = 0$ are now equivalent to the two cases.

9. As a particular example, let the cubic be $x^3 + y^3 - 1 = 0$; then $y = \sqrt[3]{1 - x^3}$, and $du = \dfrac{dx}{y^2} = \dfrac{dx}{(1 - x^3)^{\frac{2}{3}}}$; and with these values as before the differential relation $du_1 + du_2 + du_3 = 0$, and the integral relation $123 = 0$. I give a direct verification. To find x_3, y_3 the coordinates of the third intersection, we may in the equation of the cubic write $x_3, y_3 = \lambda_1 + \mu x_1, \lambda y_1 + \mu y_1$, where $\lambda + \mu = 1$, and we obtain for the determination of λ, μ the equation $\lambda_1 \cdot 1^2 2 + \mu \cdot 12^2 = 0$.

This being so, from the equation $123 = 0$ we obtain by differentiation

$$\Sigma\, (y_2 - y_3)\, dx_1 - dy_1 \cdot (x_2 - x_3) = 0,$$

the sum consisting of three terms, the second and third of them obtained from the one given by the cyclical interchange of the numbers 1, 2, 3. But we have $x_1\, dx_1 + y_1\, dy_1 = 0$, and the equation thus is

$$\Sigma\, \frac{dx_1}{y_1^2}\, \{y_1^2(y_2 - y_3) + x_1^2(x_2 - x_3)\} = 0;$$

this will reduce itself to $\Sigma\, \dfrac{dx_1}{y_1^2} = 0$, if only the three coefficients in { } are equal, that is, we ought to have

$$y_1^2(y_2 - y_3) + x_1^2(x_2 - x_3) = y_2^2(y_3 - y_1) + x_2^2(x_3 - x_1).$$

Comparing for instance the first and second terms, the equation is

$$-y_1(y_3 + y_1^2)(x_2 - x_3) + (x_1^2 + x_2^2)\, y_1(x_2 - x_3) + y_1(y_2 - y_3)(x_3 + x_2) = 0,$$

or, as this may be written,

$$-(\lambda_1 + \mu_1)(y_1^2 + y_2^2) + (\lambda_1 + \mu)x_1(x_1^2 + x_2^2) + (\lambda_1 - \mu)(x_1^2 + x_2^2)\, y_1 + (\lambda + \mu)(y_2 + y_1^2) = 0,$$

whence the whole coefficient of λ is $-x_2^2(y_1^2 + y_3^2) + x_3^2(x_1^2 + x_3^2) + (\lambda + \mu)y_3(y_1 + y_3^2) = 0$, which we see similarly the whole coefficient of μ is $-x_2^2(y_2^2 + y_3^2) + x_2^2(x_2^2 + x_3^2) + (\lambda + \mu)y_2(y_2 + y_3^2) = 0$, which is true; and similarly for the second and third terms; we have thus the required result,

$$\frac{dx_1}{y_1^2} + \frac{dx_2}{y_2^2} + \frac{dx_3}{y_3^2} = 0,$$

the integral of which is $123 = 0$. This last equation is an integral of the differential equation $du_1 + du_2 + du_3 = 0$, but it is in fact a particular integral, as it is in fact the only general integral.

[p. 115] $du_1 + du_2 + du_3 = 0$, and the integral relation $123 = 0$. This last equation is an integral of the differential equation $du_1 + du_2 + du_3 = 0$; but it is, in fact, a particular integral.

But regard one of the three points, say 3, as a fixed point, that is, the line passes through the fixed point on the cubic; then we may write $du_1 + du_2 = 0$, and the differential equation contains du_1 and du_2 only, the constants of u unaltered.

Let the variable curve be a conic; say the intersections with the cubic are 1, 2, 3, 4, 5, 6. Any five of these points determine the conic, and the displacement-relation is therefore the one displacement-relation

$$\Sigma\, du = 0,$$

which corresponds to the equation $123456 = 0$, which expresses that the six points are on a conic.

We have thus $123456 = 0$ as a particular integral of

$$du_1 + du_2 + du_3 + du_4 + du_5 + du_6 = 0.$$

If, however, we take 5 a fixed point on the cubic, then we have the same equation as the general integral of $du_1 + du_2 + du_3 + du_4 = 0$.

We have here obviously $\Sigma\, du = 0$ as a particular integral of the cubic; and the integral relation $123456 = 0$ as the general integral.

But taking also 6 a fixed point on the cubic, the equation $123456 = 0$, which expresses that the six points are on a conic, of any equation $123456 = 0$ for 4 fixed and 5 fixed point on the cubic, becomes a particular integral of $du_1 + du_2 + du_3 + du_4 = 0$.
[p. 116] determinate point, independent of the particular conic through 1, 2, 3, 4 and 5 used for the construction. We have thus a particular integral of the differential equation $du_1 + du_2 + du_3 + du_4 = 0$.

12. We might go on to the case where the variable curve is a cubic; there are here nine intersections; any eight of these do not determine the variable cubic, but they do determine the ninth intersection; and there is between the nine intersections one integral relation, and corresponding to it one displacement-relation $\Sigma du = 0$; that is, $du_1 + du_2 + \ldots + du_9 = 0$, given by the pure theorem. But as in this case the number of arbitrary constants in the conic $\Omega = 0$, the corresponding equation $\Sigma\, du = 0$ is an identity; the number of independent displacement-relations is 0.

Examples of the Affected Theorem—Fixed Curve a Conic. Art. Nos. 13 and 14.

13. The fixed curve is taken to be the conic $x^2 + y^2 - 1 = 0$, and the parametric points 1 and 2 to be the points (1, 0) and (0, 1) on this circle. The variable curve is taken to be a line, say the line $ax + by - 1 = 0$, meeting the circle in the points 3 and 4, coordinates $(x_3,\, y_3)$ and $(x_4,\, y_4)$ respectively.

Starting from the formula

$$\Sigma\, \frac{(x,\, y,\, 1)^{n-3}\, du}{012} \quad = \quad -\frac{\delta\phi_3}{\phi_3}\,\frac{\delta\phi_4}{\phi_4},$$

where the summation extends to the points 3 and 4, $(x,\, y,\, 1)^{n-3}$ is here a constant, $= 2.12$, that is, $2.(x_1x_2 + y_1y_2 - 1)$, which for the points 1, 2 in question is $= -2$. We have 012 denoting the determinant

$$012 = \begin{vmatrix} x & y & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix},$$

which is $= -x - y + 1$; and $du = \dfrac{dx}{2y}$. Also $\dfrac{\delta\phi_3}{\phi_3} = \dfrac{x_3\,\delta a + y_3\,\delta b}{ax_3 + by_3 - 1}$, is $= \dfrac{da}{a-1}$, and similarly $\dfrac{\delta\phi_4}{\phi_4}$

is $= \dfrac{db}{b-1}$. The formula thus is

$$\Sigma \frac{da}{y(x+y-1)} = -\frac{da}{a-1} + \frac{db}{b-1}.$$

The coefficients a and b are determined by means of the points 3 and 4, that is, they are functions of x_3, x_4; and considering them as thus expressed, then (inasmuch as there is no linear relation between the displacements $\dfrac{dx_3}{y_3}$ and $\dfrac{dx_4}{y_4}$ of the two arbitrary points 3 and 4 on the circle) the equation must become an identity in regard to the two lines dx_3, dx_4 respectively.

[p. 117] 14. Writing P, Q, $R = y_3 + y_4$, $x_3 - x_4$, $x_3y_4 - x_4y_3$; also L_3 and $L_4 = x_3 + y_3 - 1$ and $x_4 + y_4 - 1$ respectively, we have $a = P + R$, $b = Q + R$; and the equation is found to be

$$\frac{dx_3}{y_3L_3} + \frac{dx_4}{y_4L_4} = \frac{1}{(Q-R)(R-P)} \{(Q-R)\,dP + (R-P)\,dQ + (P-Q)\,dR\},$$

where, substituting for dy_3, dy_4 their values in terms of dx_3, dx_4, we have

$$dP,\ dQ,\ dR = \frac{1}{y_3}\,x_3\,dx_3 - \frac{1}{y_4}\,x_4\,dx_4,\ \frac{1}{y_3}\,y_3\,dx_3 - \frac{1}{y_4}\,y_4\,dx_4,\ \frac{1}{y_3}\,(x_3y_4+y_3y_4)\,dx_3 - \frac{1}{y_4}\,(x_3y_4+y_3y_4)\,dx_4,$$

and with these values, and by aid of the relations

$$Q-R,\ R-P,\ P-Q = x_3y_3 - x_4y_4,\ y_3L_4 - y_4L_3,\ -L_3 + L_4,$$

the equation is found to be

$$\frac{dx_3}{y_3L_3} + \frac{dx_4}{y_4L_4} = \frac{L_3L_4(x_3x_4 + y_3y_4 - 1)}{(x_3y_4 - x_4y_3)(y_3L_4 - y_4L_3)} \left(\frac{dx_3}{y_3L_3} + \frac{dx_4}{y_4L_4}\right);$$

viz. this will be truly identity if only

$$L_3L_4(x_3x_4 + y_3y_4 - 1) = (x_3y_4 - x_4y_3)(y_3L_4 - y_4L_3),$$

that is,

$$-x_3y_4\,L_3 + L_3L_4(x_3x_4 - 1) + (y_3y_4 + x_3y_3)(x_3 + y_3 - 1) = 0.$$

But from the values of L_3, L_4 we have $x_3 = \frac{1}{2}L_3^2 + L_3$, $x_3y_3 = L_3 + L_3$, and the coefficient of L_3L_4 is $= L_3L_4 + L_4 + L_3 + 1$, the equation is thus verified.

The example would perhaps have been more instructive if the points 1 and 2 had been left arbitrary points on the circle, but the working out would have been more difficult.

The Variable Intersections of the Two Curves—Number of Independent Integral Relations. Art. Nos. 15 to 19.

15. Suppose $n = 3$, $\delta = 0$ ($p = 1$), the fixed curve a cubic; and suppose successively $m = 1$, 2, 3, ..., the variable curve a line, conic, cubic, &c.

If $m = 1$, then two points on the cubic determine the line, and consequently the remaining intersection with the cubic; hence the number α depends.

If $m = 2$, then 5 points on the cubic determine the conic, and the remaining intersection with the cubic, hence the number α, f^2, will be a determinate point, &c.

If $m = 3$, then 8 points on the cubic determine the cubic, and consequently the remaining intersection with the cubic, hence the number α, f^3, will be a determinate point.

[p. 118] If $m = n - 3$, the variable curve is a generic functions of p, q, r, but the dependence on the parameters considered all of 1, 2, 3, &c., f which is a determinant of the variable curve.

16. Suppose next $n = 4$, $\delta = 0$ ($p = 3$); the fixed curve a general quartic; and as before suppose successively $m = 1$, 2, 3, ..., the variable curve a line, conic, cubic, &c.

If $m=1$, then two points on the quartic determine the line, and therefore the remaining two intersections; the number of integral relations is $=2$.

If $m=2$, then five points on the quartic determine the conic, and therefore the remaining three intersections; the number of integral relations is $=3$, and similarly if $m=3$, the number of integral relations is $=3$.

If $m=4$, then thirteen points on the fixed quartic do not determine the variable quartic, but they do determine the remaining three intersections. For draw through the thirteen points a no-matter-what quartic $\chi=0$; the general quartic through the thirteen points is $\chi+\alpha f=0$, and this meets the fixed quartic in the points $\chi=0$, $f=0$, that is, in the thirteen points and in three other points independent of α and thus completely determinate; and the number of integral relations is $=3$, but as α is in general for any higher value of m, the number is still $=3$.

17. Suppose $n=5$, $\delta=0$ $(p=6)$; the fixed curve a general quintic; and as before suppose $m=1, 2, 3, \ldots,$ successively.

If $m=1$, then two points on the quintic determine the line, and therefore the remaining three intersections; the number of integral relations is $=3$.

If $m=2$, then five points on the quintic determine the conic, and therefore the remaining five intersections; the number of integral relations is $=5$.

If $m=3$, then 9 points on the quintic determine the cubic, and therefore the remaining six intersections; the number of integral relations is $=6$, and so if $m=4$, or if $m=5$ or any greater number, in the first case directly, and in the other cases by consideration of the quintic $\chi+\alpha f=0$, &c., we find that the number of integral relations is always $=6$.

[p. 119] 18. The reasoning is scarcely altered in the case where the fixed curve has dps; thus considering the general case of a fixed curve n, δ, p:

If $m=1$, then $2-\delta$ points on the fixed curve (this implies $\delta\geq 2$) determine the line, and therefore the remaining $n-2\delta-(2-\delta)$, $=n-2-\delta$ intersections; the number of integral relations is $=n-2-\delta$.

If $m=2$, then $5-\delta$ points on the fixed curve (this implies $\delta\geq 5$) determine the conic, and therefore the remaining $2n-2\delta-(5-\delta)=2n-5-\delta$, and so for any value of $m\geq n-3$, and indeed for the values $n-2$ and $n-1$; here $\frac{1}{2}m(m+3)-\delta$ points on the fixed curve (this implies $\delta\geq\frac{1}{2}m(m+3)-\delta$) determine the remaining $mn-2\delta-\{\frac{1}{2}m(m+3)-\delta\}$, $=mn-\frac{1}{2}m(m+3)-\delta$ intersections. Hence the number of integral relations is $=mn-\frac{1}{2}m(m+3)-\delta$, that is, $=p-\frac{1}{2}(n-m-1)(n-m-2)$. And thus in the cases $m=n-2$ and $n-1$ the number is $=p$.

If $m=n$, then $\frac{1}{2}n(n+3)-1-\delta$ points on the fixed curve do not determine the variable curve, but they do determine the remaining p intersections, and the number of integral relations is thus $=p$; and so, for any higher value of m, the number is still $=p$.

19. The conclusion thus is

$$\begin{array}{ll} m\geq n-3, & \text{the number of integral relations} = p-\tfrac{1}{2}(n-m-1)(n-m-2),\\ m=n \text{ or } >n-3, & \quad ,, \qquad = p. \end{array}$$

The integral equations spoken of throughout are of course independent relations.

The Variable Intersections of the Two Curves. Number of Independent Displacement Relations given by the Pure Theorem. Art. No. 20.

20. The number of displacement-relations given by the pure theorem is $=$ number of constants in minor function $(x, y, z)^{n-3}$, which equated to zero represents a curve through the dps, viz. this is always

$$\tfrac{1}{2}(n-1)(n-2)-\delta, \ =p.$$

But for $m\geq n-2$, these relations are not independent. For instance, for $n=4$, $\delta=0$, $m=1$, the displacement-relations are

$$\Sigma\,(x, y, z)^2\,du=0, \text{ that is, } \Sigma x\,du=0,\ \Sigma y\,du=0,\ \Sigma z\,du=0,$$

and conversely from these last equations we have $\Sigma\,(x,\,y,\,z)^2\,du = 0$. But in this case the variable curve is a line meeting the quartic in four points; the equation $\Sigma x\,du = 0$ is satisfied for each of the intersections of the line with the quartic, and the corresponding equation $\Sigma\,(x,\,y,\,z)^2\,du = 0$ is an identity. Hence the number of independent displacement-relations is $3 - 1, = 2$.

So for $n = 5$, $\delta = 0$, $m = 1$, the displacement-relations are

$$\Sigma\,(x,\,y,\,z)^2\,du = 0, \text{ that is, } \Sigma(x^2,\,y^2,\,z^2,\,yz,\,zx,\,xy)\,du = 0,$$

and so on for $n = 6$, $\delta = 0$, $m = 1$, the displacement-relations are

$$\Sigma\,(x,\,y,\,z)^3\,du = 0.$$

[p. 120] and six equations give conversely $\Sigma\,(x,\,y,\,z)^2\,du = 0$, in particular they give $\Sigma x\,du = 0$, $\Sigma y\,du = 0$, $\Sigma z\,du = 0$, $\Sigma xy\,du = 0$. But if $(x,\,y,\,z)$ denote any of the intersections of the line with the curve, then for each of the five intersections $(x,\,y,\,z) = 0$, that is, the fast-mentioned three equations are identities. Hence the number of independent displacement-relations is $6-3$, $= 3$.

So for $n = 5$, $\delta = 0$, $m = 2$. Here if the variable curve is $\phi = (a, \dots\ )(x,\,y,\,z)^2$, then taking $(x,\,y,\,z) = (x_1,\,y_1,\,z_1)$, $(x_2,\,y_2,\,z_2)$, $\dots$, the corresponding equation $\Sigma\,(x,\,y,\,z)^2\,du = 0$ is an identity; the number of independent displacement-relations is $6 - 1, = 5$.

The reasoning is the same when δ is not $= 0$, and we see generally that for $m \geq n - 2$,

$$m \geq n - 3, \quad \text{the number of independent displacement-relations} \\ = p - \tfrac{1}{2}(n - m - 1)(n - m - 2);$$

while for

$$m = n \text{ or } > n - 2, \text{ the number is } = p;$$

since in this case the relations given by the theorem are all of them independent. It thus appears *à posteriori*, that in every case the number of independent displacement-relations given by the pure theorem is equal to the number of independent integral relations.

As to the dps of the Fixed Curve. Art. No. 21.

21. I conclude with a general remark applicable to the whole of the three chapters. There is no necessity to attend to all or indeed to any of the dps of the fixed curve. Suppose that the fixed curve has $\delta + \delta'$ dps, where δ, δ' may be either of them or each $= 0$, and attend only to the δ dps, the δ' dps being wholly disregarded, and accordingly let the expression the dps mean as before the δ dps of the fixed curve. No attention at all is required: only, if δ' be not $= 0$, then $p = \frac{1}{2}(n - 1)(n - 2) - \delta - \delta'$ will no longer be the deficiency. To obtain the best theorem we use all the $\delta + \delta'$ dps; but disregarding the δ' dps, we obtain theorems, as for a curve with δ dps, which are true and may frequently be useful either in their original form or with simplifications introduced therein by afterwards taking account of the δ' dps.

Chapter II. Proof of Abel's Theorem.

Preparation. Art. Nos. 22 and 23.

22. Starting from the equation $\phi = (a, \dots\ )(x,\,y,\,z)^m = 0$ of the variable curve, we have

$$\frac{d\phi}{dx}\,dx + \frac{d\phi}{dy}\,dy + \frac{d\phi}{dz}\,dz + \delta\phi = 0,$$

$$\frac{d\phi}{dx}\,x + \frac{d\phi}{dy}\,y + \frac{d\phi}{dz}\,z = 0,$$

where $\delta\phi = (da, \ldots\ )(x,\ y,\ z)^m$. Let τ denote an arbitrary linear function, $= ax + by + cz$; multiply the two equations by $ax+by+cz$, $=\tau$, and $-(x\,dz + b\,dy + c\,dz)$, $= -d\tau$ respectively, and add: we obtain

$$(y\,dz - z\,dy)\left(a\frac{d\phi}{dz} - c\frac{d\phi}{dy}\right) + (z\,dx - x\,dz)\left(b\frac{d\phi}{dx} - a\frac{d\phi}{dz}\right) + (x\,dy - y\,dx)\left(a\frac{d\phi}{dy} - b\frac{d\phi}{dx}\right) + \tau\,\delta\phi = 0;$$

introducing du, this becomes

$$du\left[\frac{df}{dx}\left(b\frac{d\phi}{dz} - c\frac{d\phi}{dy}\right) + \frac{df}{dy}\left(c\frac{d\phi}{dx} - a\frac{d\phi}{dz}\right) + \frac{df}{dz}\left(a\frac{d\phi}{dy} - b\frac{d\phi}{dx}\right)\right] + \tau\,\delta\phi = 0,$$

or observing that a, b, c are the differential coefficients $\frac{d\tau}{dx}, \frac{d\tau}{dy}, \frac{d\tau}{dz}$, the term in [ ] is $J(f,\ \tau,\ \phi)$, or say $J(\phi,\ f,\ \tau)$, and the equation is

$$du\,J(\phi,\ f,\ \tau) + \tau\,\delta\phi = 0,$$

where $J(\phi,\ f,\ \tau)$ is the Jacobian, or functional determinant

$$\begin{vmatrix} \frac{d\phi}{dx} & \frac{d\phi}{dy} & \frac{d\phi}{dz} \\ \frac{df}{dx} & \frac{df}{dy} & \frac{df}{dz} \\ \frac{d\tau}{dx} & \frac{d\tau}{dy} & \frac{d\tau}{dz} \end{vmatrix},\ = \frac{d(\phi,\ f,\ \tau)}{d(x,\ y,\ z)};$$

and we hence have

$$du = -\frac{\tau\,\delta\phi}{J(\phi,\ f,\ \tau)}.$$

23. The two theorems thus become

$$\Sigma\,(x,\ y,\ z)^{n-3}\,\frac{\tau\,\delta\phi}{J(\phi,\ f,\ \tau)} = 0,$$

$$\Sigma\,(x,\ y,\ z)^{n-3}\,\frac{-\,\tau\,\delta\phi}{012\,J(\phi,\ f,\ \tau)} = -\frac{\delta\phi_1}{\phi_1}\,\frac{\delta\phi_2}{\phi_2}.$$

But further, if in the first equation we retain τ, while in the second equation we retain τ, using it to denote the linear function 012, the equations become

$$\Sigma\,(x,\ y,\ z)^{n-3}\,\frac{\delta\phi}{J(\phi,\ f)} = 0,$$

$$\Sigma\,(x,\ y,\ z)^{n-3}\,\frac{-\,\delta\phi}{J(\phi,\ f,\ \tau)} = -\frac{\delta\phi_1}{\phi_1}\,\frac{\delta\phi_2}{\phi_2};$$

where, in the first equation, $J(\phi,\ f)$ denotes the Jacobian

$$\begin{vmatrix} \frac{d\phi}{dx} & \frac{d\phi}{dy} \\ \frac{df}{dx} & \frac{df}{dy} \end{vmatrix},\ = \frac{d(\phi,\ f)}{d(x,\ y)}.$$

[p. 122] In these equations the only differential symbol is the δ affecting the coefficients a, b, $\ldots$ of ϕ, ϕ_1, ϕ_2; the equations are, in respect to the coordinates $(x,\ y,\ z)$ of the several variable intersections, but they are mere algebraical equations, which are in fact given by Jacobi's

Fraction-theorem about to be explained. But for the further reduction of the affected theorem I interpose the next three articles.

The Coordinates (ρ, σ, τ). Art. Nos. 24 to 26.

24. The letter τ has just been used to denote the determinant 012; there is often occasion to consider three points 1, 2, 3 coordinates (x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3) respectively; and then writing ρ, σ, τ to denote the determinants 023, 031, 012 respectively, and Δ the determinant 123, we have identically

$$\begin{aligned}\Delta x &= x_1\rho + x_2\sigma + x_3\tau,\\ \Delta y &= y_1\rho + y_2\sigma + y_3\tau,\\ \Delta z &= z_1\rho + z_2\sigma + z_3\tau,\end{aligned}$$

which equations, regarding therein the point 0, coordinates (x, y, z), as a current point, are in fact equations for transformation from the coordinates (x, y, z) to the coordinates ρ, σ, τ, belonging to the triangle of reference 123. The points 1 and 2 have been already taken to be on the fixed curve, and it will be assumed that 3 is also a point on this curve.

25. If the function f, which equated to zero gives the equation of the fixed curve, be transformed to the new coordinates (ρ, σ, τ), the coefficients of the transformed function are polar functions, each divided by ∇^n, viz. the coefficient of ρ^n is $\dfrac{1}{\nabla^n}\,1^n$, which by reason that 1 is a point on the curve is $= 0$ (and similarly the coefficients of σ^n and of τ^n are each $= 0$); the coefficient of $\rho^{n-1}\sigma$ is $\dfrac{1}{\nabla^n}\,n.\,1^{n-1}\,2$; that of $\rho^{n-1}\tau$ is $\dfrac{1}{\nabla^n}\,n.\,1^{n-1}\,3$; that of $\rho^{n-2}\sigma^2$ is $\dfrac{1}{\nabla^n}\,\tfrac{1}{2}n(n-1).\,1^{n-2}\,2^2$; and so in other cases. I write this in the form

$$f = \frac{1}{\nabla^n}\,(1^n, \ldots\ )(\rho, \sigma, \tau)^n;$$

or we might also use the symbolic form

$$f = \frac{1}{\nabla^n}\,(\rho 1 + \sigma 2 + \tau 3)^n.$$

[p. 123] The terms independent of τ contain, it is clear, the factor $\rho\sigma$, and separating these terms from the others, the equation may be written

$$f = \frac{1}{\nabla^n}\,\rho\sigma\,(n.\,1^{n-1}\,2, \ldots\ )(\rho, \sigma)^{n-2} + \text{terms involving } \tau.$$

26. The equations $\tau = 0$, $(\ldots\ )(\rho, \sigma)^{n-2} = 0$, determine the residues of the points 1, 2, and hence the major function $(x, y, z)^{n-3}$, expressed in terms of ρ, σ, τ, and writing therein $\tau = 0$, must reduce itself to $(\ldots\ )(\rho, \sigma)^{n-2}$ multiplied by a constant factor which can be found to be $= \dfrac{1}{\nabla^{n-1}}$: for taking the current point at 1 we have $(\rho, \sigma, \tau) = (\Delta, 0, 0)$, and the corresponding value of the major function is thus equal, and in this case $\dfrac{1}{\nabla^{n-1}}\,n.\,1^{n-1}\,2$, $= n.\,1^{n-1}\,2$, as it ought to be. We have thus

$$(x, y, z)^{n-3} = \frac{1}{\nabla^{n-1}}\,(n.\,1^{n-1}\,2, \ldots\ )(\rho, \sigma)^{n-2} + \text{terms involving } \tau;$$

and we hence see that, for $\tau = 0$,

$$(x, y, z)^{n-3} = \frac{\Delta f}{\rho\sigma}.$$

an equation which will be useful.

The Preparation for the Affected Theorem resumed. Art. No. 27.

27. In the affected theorem instead of (x, y, z) we introduce the new coordinates (ρ, σ, τ). We have

$$J(\phi, f, \tau) = \frac{d(\phi, f, \tau)}{d(\rho, \sigma, \tau)} \frac{d(\rho, \sigma, \tau)}{d(x, y, z)},$$

where the first factor is $= \dfrac{d(\phi, f)}{d(\rho, \sigma)}$, say that this is $J(\phi, f)$, the Jacobian in regard to ρ, σ: and the second factor is at once found to be $= \Delta!$. We have consequently

$$\frac{1}{J(\phi, f, \tau)} = \frac{1}{\Delta!\, J(\phi, f)},$$

and the equation for the affected theorem becomes

$$\Sigma\, (x, y, z)^{n-3} \frac{\delta\phi}{J(\phi, f)} = -\,\Delta! \left(-\frac{\delta\phi_1}{\phi_1} \frac{\delta\phi_2}{\phi_2} \right),$$

where $(x, y, z)^{n-3}$ is to be regarded as standing for its value in terms of (ρ, σ, τ).

Jacobi's Fraction-Theorem. Art. Nos. 28 to 31.

28. This is the extension of a well-known theorem, which, in a somewhat disguised form, may be thus written: viz. if U be any rational and integral function $(x, 1)^m$, then we have

$$\frac{1}{U} = \Sigma \frac{1}{x - x'} \frac{1}{J(U)},$$

[p. 124] or, introducing an arbitrary constant A by the equation $AU = X$, say this is

$$\frac{A}{X} \left(= \frac{1}{U} \right) = \Sigma \frac{1}{x - x'} \frac{1}{J(U)},$$

where U' is the same function (of x') that U is of x; $J(U)$, $= \dfrac{dU'}{dx'}$, is the Jacobian of U', and the summation extends to all roots x' of the equation $U' = 0$: obviously this is nothing else than the formula for the decomposition of $\dfrac{1}{U}$ into simple fractions.

29. Take now $U = (x, y, 1)^m$, $V = (x, y, 1)^p$ functions of x, y of the degrees m and n respectively, and assume

$$\begin{aligned} AU + BV &= X, \quad \text{a function of } (x, 1)^{mn}, \\ CU + DV &= Y, \quad \text{,, of } (y, 1)^{mn}; \end{aligned}$$

viz. let $X = 0$ and $Y = 0$ be the equations obtained by elimination from $U = 0$ and $V = 0$ of the y and the x respectively. The forms are

$$\begin{aligned} A &= (x, y', 1)^{mn-m}, \quad & B &= (x, y', 1)^{mn-n}, \\ C &= (x^{m-1}, y, 1)^{mn-m}, \quad & D &= (x^{m-1}, y, 1)^{mn-n}; \end{aligned}$$

where these equations denote the kind of them; that is A is a rational and integral function of the degree $mn - m$ in x and y jointly, but only of the degree $mn - 1$ in y; and so for the other equations. It follows that

$$AD - BC = (x^{mn-1}, y^{mn-1}, 1)^{mn-mn-m}.$$

The theorem now is

$$\frac{AD - BC}{XY} = \Sigma \frac{1}{x - x' \cdot y - y'} \frac{1}{J(U', V')},$$

where U', V' are the same functions of (x', y') that U, V are of (x, y); $J(U', V')$ is the Jacobian $\dfrac{d(U', V')}{d(x', y')}$, $= 0$; and the summation extends to all the simultaneous roots x', y' of the equations $U = 0$, $V = 0$.

30. For the proof, observe that $AD - BC$ is a sum of terms of the form $\alpha x^\alpha y^\beta$ where α and β are each of them at most $= mn - 1$; hence X being of the degree mn we have $\dfrac{x^\alpha}{X}$ a sum of fractions $\dfrac{L_y}{x - x'}$, where x' is any root of $X = 0$; and similarly $\dfrac{y^\beta}{Y}$ a sum of fractions $\dfrac{M_y}{y - y'}$, where y' is any root of $Y = 0$; multiplying the two expressions and taking into account the several terms $\lambda x^\alpha y^\beta$ of $AD - BC$ we obtain a formula

$$\frac{AD - BC}{XY} = \Sigma \frac{K}{x - x' \cdot y - y'},$$

where the summation extends to all the combinations of the mn values of x' with the mn values of y. But such a formula existing, the coefficients K may be determined.
[p. 125] in the usual manner, viz. multiplying by XY and then writing $x = x'$, $y = y'$, there is on the right-hand only one term which does not vanish, and we find

$$(AD - BC)_{xy'} = K\left(\frac{X}{x - x'}\right)_x \left(\frac{Y}{y - y'}\right)_{y'}, \quad = \; K\left(\frac{dX}{dx}\,\frac{dY}{dy}\right)_{xy'};$$

where the factor which multiplies K does not vanish.

We distinguish the cases where (x', y') are corresponding or non-corresponding roots of $X = 0$, $Y = 0$, viz. corresponding roots are those for which $U = 0$, $V = 0$, but for non-corresponding roots these equations do not hold good; there are obviously mn pairs of corresponding roots.

In the latter case $(AD - BC)\,U = DX - BY$, $(AD - BC)\,V = -CX + AY$, and since for these values have $X = 0$, $Y = 0$, the foregoing equation for K gives then $K = 0$.

31. The formula thus is

$$\frac{AD - BC}{XY} = \Sigma \frac{K}{x - x' \cdot y - y'},$$

where the summation extends to corresponding roots only; or, for which we have $U = 0$, $V = 0$.